ART FIBER SYSTEMS CHALLENGE · SCHUBKI NATIONAL AI HACKATHON 2026

AFS Spectroscopy Assistant

A customer enquiry goes in. A cited first technical assessment comes out.

Team Helios

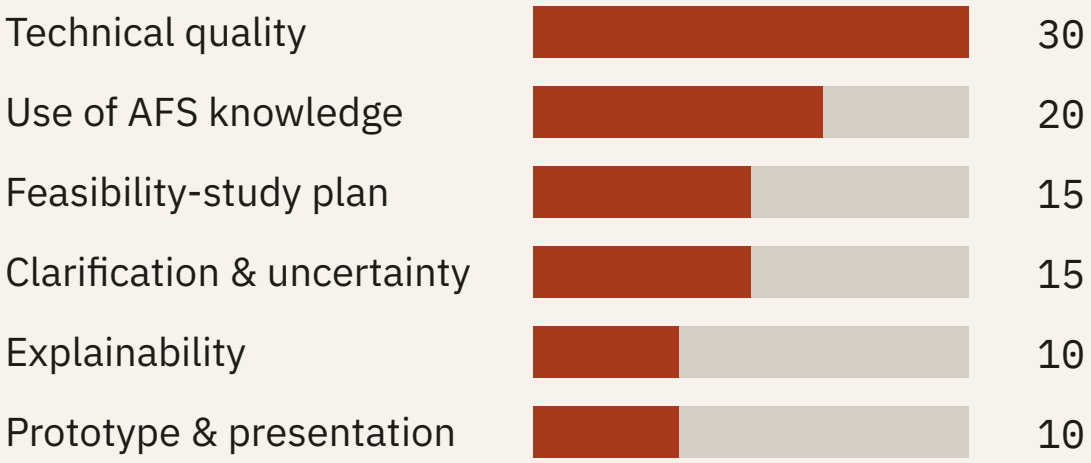
TEAM_MEMBERS_PENDING

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Help an AFS expert answer a new customer

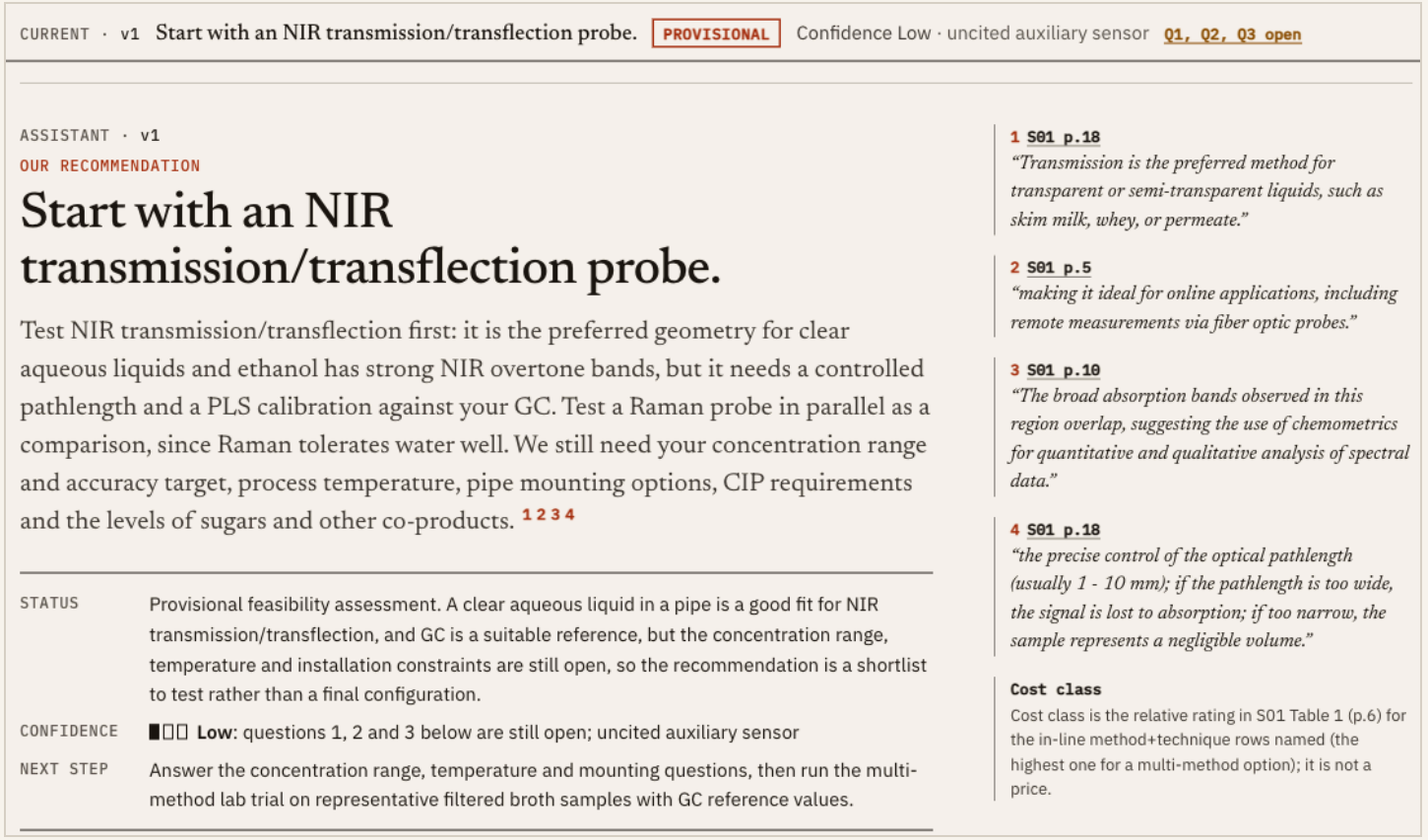
- A customer describes what they want to measure in their process. **Which method and probe should they test first?**
- The answer must rest on the **AFS documents**. Nothing invented: no prices, accuracy figures or fiber lengths.
- It should **ask the right questions** and **propose a first feasibility test**.

HOW IT IS JUDGED (WEIGHT %)



THE JURY ALSO ASKED Always make a proposal. Search the literature when the AFS documents don’t cover it. Handle setups that combine several methods.

A chat assistant that writes the first assessment



CURRENT · v1 Start with an NIR transmission/transflection probe. PROVISIONAL Confidence Low · uncited auxiliary sensor Q1, Q2, Q3 open

ASSISTANT · v1

OUR RECOMMENDATION

Start with an NIR transmission/transflection probe.

Test NIR transmission/transflection first: it is the preferred geometry for clear aqueous liquids and ethanol has strong NIR overtone bands, but it needs a controlled pathlength and a PLS calibration against your GC. Test a Raman probe in parallel as a comparison, since Raman tolerates water well. We still need your concentration range and accuracy target, process temperature, pipe mounting options, CIP requirements and the levels of sugars and other co-products. 1 2 3 4

STATUS	Provisional feasibility assessment. A clear aqueous liquid in a pipe is a good fit for NIR transmission/transflection, and GC is a suitable reference, but the concentration range, temperature and installation constraints are still open, so the recommendation is a shortlist to test rather than a final configuration.
CONFIDENCE	■□□ **Low:** questions 1, 2 and 3 below are still open; uncited auxiliary sensor
NEXT STEP	Answer the concentration range, temperature and mounting questions, then run the multi-method lab trial on representative filtered broth samples with GC reference values.

1 S01 p.18
"Transmission is the preferred method for transparent or semi-transparent liquids, such as skim milk, whey, or permeate."

2 S01 p.5
"making it ideal for online applications, including remote measurements via fiber optic probes."

3 S01 p.10
"The broad absorption bands observed in this region overlap, suggesting the use of chemometrics for quantitative and qualitative analysis of spectral data."

4 S01 p.18
"the precise control of the optical pathlength (usually 1 - 10 mm); if the pathlength is too wide, the signal is lost to absorption; if too narrow, the sample represents a negligible volume."

Cost class
Cost class is the relative rating in S01 Table 1 (p.6) for the in-line method+technique rows named (the highest one for a multi-method option); it is not a price.

A live answer. Each numbered source on the right is the exact AFS quote with its page.

Recommendation
What to test first, with a decision status and a confidence level.

Three options
Simple, recommended, extended. Cost class from AFS Table 1, never a price.

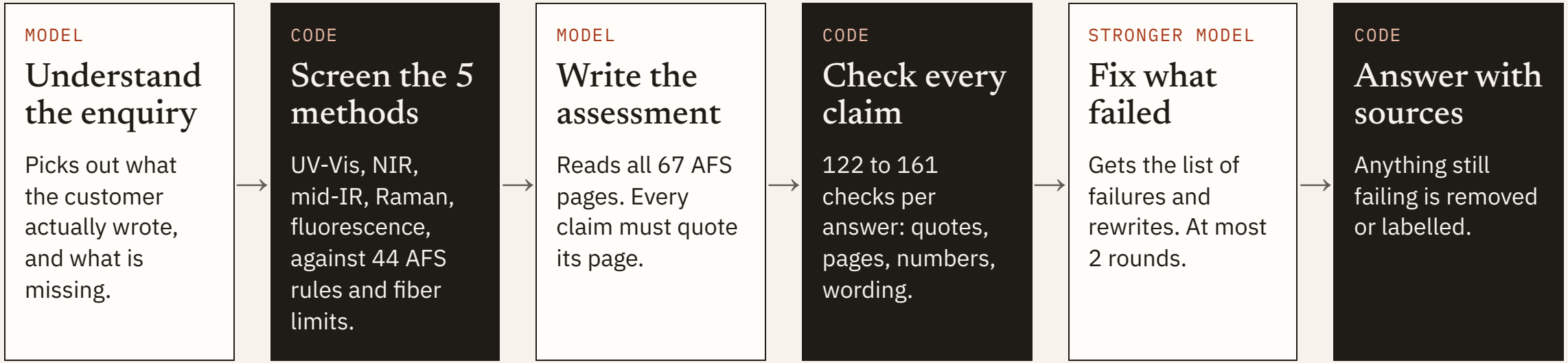
Setup sketch
Process, probe, fiber, spectrometer.

Questions for the customer
The ones that could change the choice.

Chat
Ask a follow-up or add a fact: the answer updates.

Screenshot: crop of docs/video/frames/09_recommendation.png (live run in the web chat) · README.md "What it does"

Models write. Code checks.



☐ language model ■ code: same input, same result

> The literature is searched only for gaps the AFS documents leave open. Those quotes are shown apart and marked **secondary**, not AFS material.

> The same steps run for every enquiry. Nothing is written for a specific case.

The AFS files become checkable knowledge

WHAT AFS GAVE US

- **Lab report, slide deck, flyer**
  PDFs (S01, S02, S05); slide images typed in by hand
- **Two research papers**
  kidney tissue study (S03), mid-IR + fluorescence (S04)
- **Guidance document**
  answer structure and method overview (REF)
- **Table 1 spreadsheet**
  methods compared, with a cost rating
- **A laboratory web page**
  saved as text (S06)

→

WHAT THE CODE BUILDS FROM THEM

67 pages of text

The model reads them. The checker compares every quote against them.

44 rules

Each backed by a word-for-word AFS quote. Example: silica fibers stop near 2.4 µm, so they cannot carry mid-IR (fiber chart, S02 p.6).

19 Table 1 rows

Method and probe type, with the cost class: low, average or high (S01 p.6).

7 fiber families

The wavelength range each one carries (S02 p.6).

Same files in, same files out. A check confirms all 77 rule quotes are on their page.

Four checks on every claim

1

The quote is on the page

Word for word. A changed number or a dropped “not” fails.

2

The page fits the claim

A similar application elsewhere is not proof for this customer.

3

Numbers have a source

Only from the quote or the customer’s own text.

4

No promises

No prices, no guarantees, no approval wording.

IF A CHECK FAILS

The failures go to the stronger fixer model, which rewrites.

↓ still failing?

The claim is removed, or labelled so the reader sees it.

120/120 citations verified in the six held-out answers

WHAT THE READER SEES NEXT TO EACH CLAIM

`AFS source`

backed by a checked AFS quote.

`precedent only`

a similar published use: a reason to test, not proof.

`engineering judgement, no source`

our reasoning, clearly marked as such.

Each job goes to the model that does it best

ROLE	MODEL	JOB
Reader	`Gemini 3.5 Flash-Lite`	Understands the enquiry; writes literature searches.
Writer	`GLM-5.3-Flash`	Writes the first draft of the assessment.
Fixer	`Gemini 3.8 Flash`	Rewrites, only when checks fail.
Backup	`every step`	Retries on a different model if a reply breaks.

In all six held-out cases the checks caught problems in the first draft, so each answer shown is the fixer's rewrite.

WHY THIS MIX

We tested 7 setups blind on 8 practice cases, scored by 4 judge agents.

SETUP	QUALITY /5	TIME
Best single model	4.15	259 s
Our old setup	4.08	504 s
This mix	**4.06**	**84 s**

Quality on par, **3–6× faster**, about **$0.03** per case.

The six held-out questions, answered by the agent

	THE CUSTOMER ASKS	WHAT WAS TRICKY	THE AGENT'S VERDICT
H01	New powder blend: can we reuse our NIR calibration if we add Raman?	A calibration only fits the mixture it was built on.	PREMISE NOT SUPPORTED Rebuild the NIR calibration on the new blend; test Raman separately.
H02	Clear whey and opaque milk concentrate: what should we test first?	Two very different streams.	PROVISIONAL NIR: transmission/transflection for the whey, diffuse reflection for the concentrate.
H03	Can our fluorescence spectrometer report glucose and viable biomass?	Glucose is not fluorescent; no marker named.	PREMISE NOT SUPPORTED Test Raman first, NIR as the comparison.
H04	Reuse a silica cable for MIR ATR on a petroleum line?	Silica fibers cannot carry mid-IR light.	PREMISE NOT SUPPORTED Test MIR ATR, with a CIR or PIR fiber.
H05	A liquid goes off-spec. Give us the exact sensor now.	Contaminant and quality variable unknown.	INSUFFICIENT INFORMATION Lab trial first: Raman and mid-IR ATR on normal and off-spec samples.
H06	Does the kidney study prove our sensor is ready for live surgery?	It is an ex vivo proof of concept; no own data.	PREMISE NOT SUPPORTED Ex vivo NIR test on own tissue, with histopathology as reference.

Run once on the frozen system; outputs not edited. An independent review found all six reach the right core verdict.

Reuse of a silica link for petroleum MIR ATR

Premise not supported: the sources do not support the customer's premise as stated; the assessment below is provisional.

The concept cannot be approved as stated: silica fibers cannot carry mid-IR light, so MIR ATR needs a CIR or PIR fiber instead of the existing cable. MIR ATR itself is worth testing first, with a published precedent for water in petroleum. Open items: the MIR link length to the other building, the sample state, and the required accuracy.

Proposed now: the recommended option (basis: supplied AFS sources): MIR, ATR probe as primary candidate [S02 p.13; S01 p.9].

CLARIFICATION QUESTIONS

1. What is the actual link length and routing between the petroleum line and the analyzer building?
2. What is the physical state and condition of the petroleum at the measurement point (liquid, viscous, opaque, temperature, fouling)?
3. What water-content range, accuracy and response time does the investigation require?

The six held-out questions, run once

6/6 questions answered in full	**120**/120 citations verified on the cited page	**0** blocking check failures left in any answer
1.5–3 min per question	**$0.29** for all six questions together	**1.4–6.3** s per chat follow-up (10 tested)

890 automated tests pass, with no model calls.

Next steps

NEXT

A short customer reply by default

Full answers are about 3,000 words; the short view built by code is 560–698.

NEXT

A second checker: does the quote really support the sentence?

Today a quote is verified when it is on the right page.

NEXT

Confidence that moves with what is known

Today most options say "Low" (45 of 55 in our review).

NEXT

Fewer questions when the enquiry is already complete

A complete test enquiry still got 4–5 questions.

NEXT

Faster runs

The fixing step takes 46–72 % of each run.

THANK YOU

Team Helios

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Demo video: 2:53